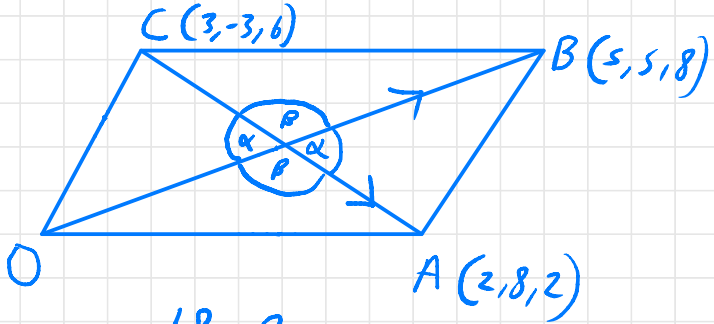


$180 - \beta = \alpha$

$$\overrightarrow{OB} = \begin{pmatrix} 5 \\ 5 \\ 8 \end{pmatrix}$$

$$\overrightarrow{CA} = \begin{pmatrix} 2 \\ 8 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} -1 \\ 11 \\ -4 \end{pmatrix}$$

$$\cos\theta = \frac{\begin{pmatrix} 5 \\ 5 \\ 8 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 11 \\ -4 \end{pmatrix}}{\sqrt{5^2+5^2+8^2}\ \sqrt{1^2+11^2+4^2}} = \frac{-5+55-32}{\sqrt{114}\ \sqrt{138}} = \frac{18}{6\sqrt{437}}$$

81.749

81.7

$y = x^2 + 2x + 5$

$y = (x+1)^2 - 1 + 5$

$y - 4 = (x+1)^2$

$\sqrt{y-4} = x+1$

$x = \sqrt{y-4} - 1$

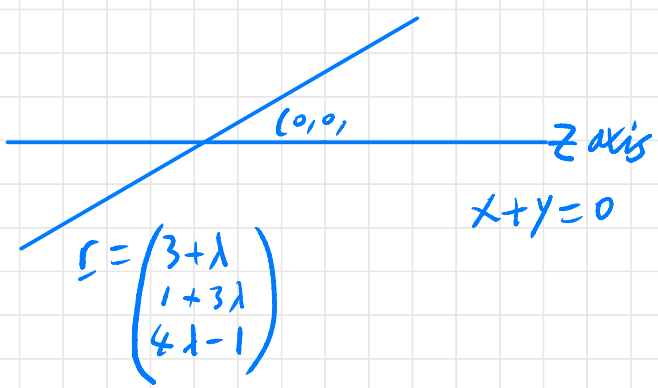
$f^{-1}(x) = (x-4)^{1/2} - 1$

$f^{-1}(x) = g(x)$

$g'(x) = \frac{1}{2}(x-4)^{-1/2}$

$\frac{1}{2}(4)^{-1/2} = \frac{1}{4}$

$y = -4x + c \qquad (8, 1)$



$\underline{r} = \begin{pmatrix} 3+\lambda \\ 1+3\lambda \\ 4\lambda - 1 \end{pmatrix}$

$$r \cdot \begin{pmatrix} 1 \\ 1 \\ -5 \end{pmatrix} = 19 \qquad \begin{matrix} 1 \\ 3 \\ 4 \end{matrix}$$

$$\cos\theta = \left| \frac{\begin{matrix} 1 \\ 1 \\ -5 \end{matrix} \cdot \begin{matrix} 1 \\ 3 \\ 4 \end{matrix}}{\sqrt{27}\sqrt{26}} \right| = \frac{16}{\sqrt{27}\sqrt{26}}$$